
DIGITAL FORENSICS · ENTERPRISE PLATFORM

Digital_V3

The Evidence Pipeline That
Holds Up in Court

Immutable blockchain logging • Deep binary metadata extraction
Role-based access control • Court-ready certificate generation

"In a court of law, digital evidence is only as powerful as its provenance. Digital_V3 makes provenance unbreakable."

Executive Overview

Digital forensics investigations live or die on two pillars: the **integrity** of the evidence and the **clarity** of its custody history. A single gap in the chain — one unlogged file access, one unverified hash — is all the defense needs to cast reasonable doubt.

Digital_V3 was engineered to eliminate those gaps entirely. It is a unified, forensic-grade evidence management platform that ingests raw seized files and immediately wraps them in a cryptographic and procedural framework that is — by design — impossible to quietly undo.

4

Independent Security Layers

100%

Blockchain-Anchored Evidence

Zero

Silent Tampering Possible

The Problem It Solves

Modern investigations generate thousands of digital artifacts. Managing that volume while preserving court-admissible integrity has historically required multiple disconnected tools, manual custody logs, and significant analyst overhead. The risks compound:

- **Tampered files with no detection** — without hash verification, evidence can be silently modified after seizure.
- **Incomplete audit trails** — gaps in custody logs are regularly exploited in court to challenge admissibility.
- **Hidden file origins** — images and documents carry embedded metadata (GPS coordinates, software signatures, AI-generation markers) that standard tools miss.
- **Unauthorized access** — without role-based controls, sensitive evidence is exposed to personnel who have no business handling it.

Digital_V3 addresses every one of these risks through a single, cohesive platform — eliminating the integration overhead of juggling separate tools while delivering a higher standard of forensic certainty than any patchwork solution can.

Core Features

3.1 Immutable Blockchain Evidence Ledger

Every file that enters Digital_V3 is immediately fingerprinted with both **SHA-256** and **MD5** hashes, then registered as a block on a custom-built Python blockchain. This is not a database record that can be quietly edited. Each block contains:

Action Type & Timestamp

What happened and precisely when — upload, analysis, transfer, or verification.

Responsible User ID

The identity of every operator who triggered the action, permanently recorded.

Cryptographic File Hash

The file's unique SHA-256 fingerprint, proving it has not changed since the event.

Previous Block Hash

Chains each event to the one before it. Altering any block breaks every block after it.

3.2 Deep Binary Metadata Extraction (The Forensic Microscope)

A file's visible content is only part of the story. The `metadata_extractor` module dissects files at the binary level to surface what was never meant to be seen:

EXIF & GPS Coordinates

[Images / Video](#)

Extracts camera make, model, GPS latitude/longitude, and capture timestamp directly from image headers — placing a photo at a specific location and time.

C2PA & JUMBF Manifest Reading

[Provenance Standard](#)

Reads the emerging Coalition for Content Provenance and Authenticity manifests — the new industry standard for tracing whether media was AI-generated or edited, adopted by Adobe, Microsoft, and major camera manufacturers.

Software Signature & AI-Generation Detection

[Manipulation Score](#)

Detects fingerprints from tools like Photoshop, Lightroom, DALL-E, and Midjourney embedded in file metadata. Produces a mathematical manipulation score — objective evidence of alteration for court presentation.

Raw Binary Header & MIME Type Forensics

[All File Types](#)

Reads raw file headers to confirm true MIME type — catching files disguised by extension renaming. Also scans PDFs, archives, and video files for embedded metadata and suspicious payloads.

3.3 Complete Chain of Custody Tracking

Every interaction with every piece of evidence — from the moment of first upload through every analyst hand-off to final court submission — is recorded in both the relational database *and* the blockchain. The **Custody Log** captures:

Event	Recorded Detail
Evidence Upload	Uploader identity, timestamp, hash, file type, case ID
Analyst Access	Which analyst opened the file, time, IP, duration
Transfer	Source operator, recipient operator, reason, hash-verified
Verification	Hash comparison results, verification officer, timestamp
Export / Print	Certificate generation events, recipient, format

Security Architecture

4.1 Role-Based Access Control (RBAC)

Digital_V3 enforces a strict three-tier access hierarchy. No user can exceed the permissions of their assigned role. Every API endpoint is decorator-protected:

Admin

Full system access.
User creation, case assignment, audit review, and system configuration.

Investigator

Case creation and management. Evidence upload and transfer.
Read access to assigned cases only.

Analyst

Read-only access to assigned evidence.
Annotation and reporting.
No modification rights.

4.2 JWT Authentication & Rate Limiting

All sessions are managed via cryptographically-signed **JSON Web Tokens (JWT)** — stateless, expiry-enforced, and tamper-evident. The API layer enforces rate limiting at the endpoint level, preventing automated brute-force and enumeration attacks on evidence collections.

Court-Ready Certificate Generation

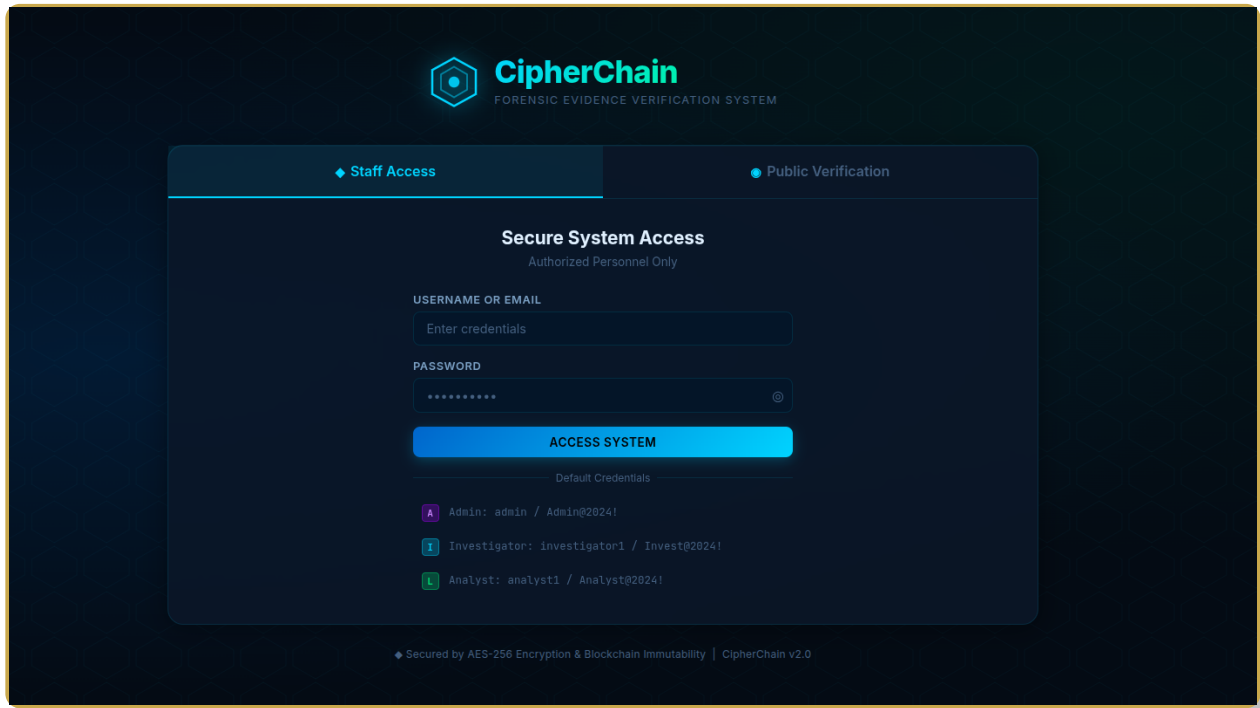
When evidence reaches the final stage of an investigation, Digital_V3 generates a **forensic verification certificate** — a cryptographically-anchored PDF document containing the complete chain of custody timeline, hash verification proofs, blockchain block references, and analyst sign-off records. This document is designed from the ground up to satisfy the evidentiary standards expected by courts and legal proceedings.

Certificate Contents: File identity & verified hash values · Blockchain block index & previous hash · Complete access and transfer timeline · Software signature analysis report · GPS and EXIF provenance summary · Issuing officer digital signature

Platform Interface Gallery

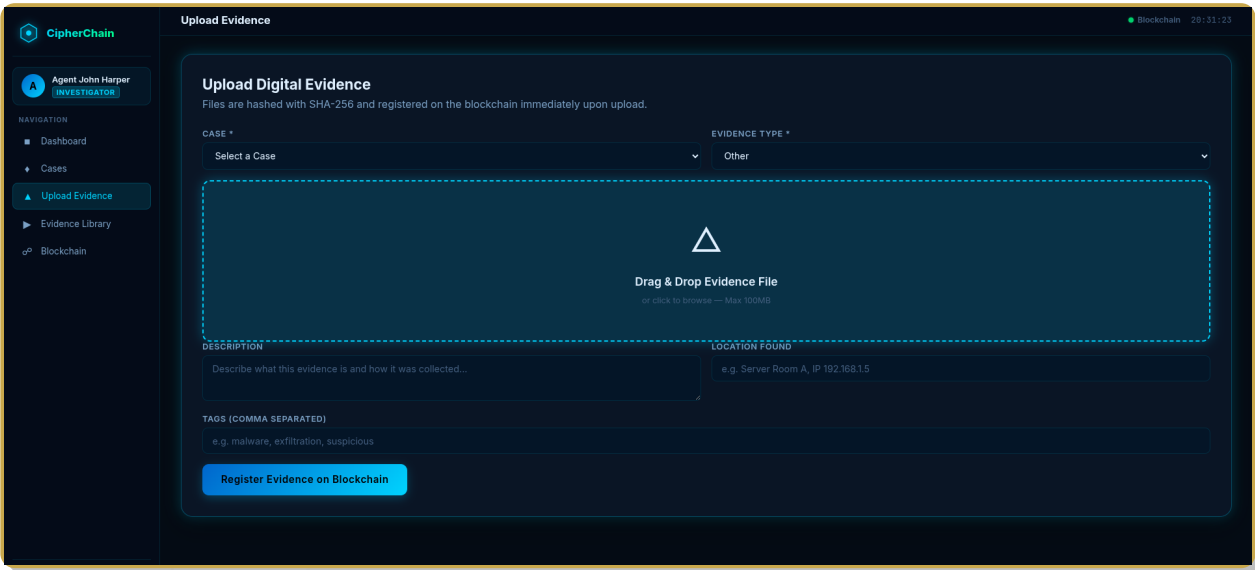
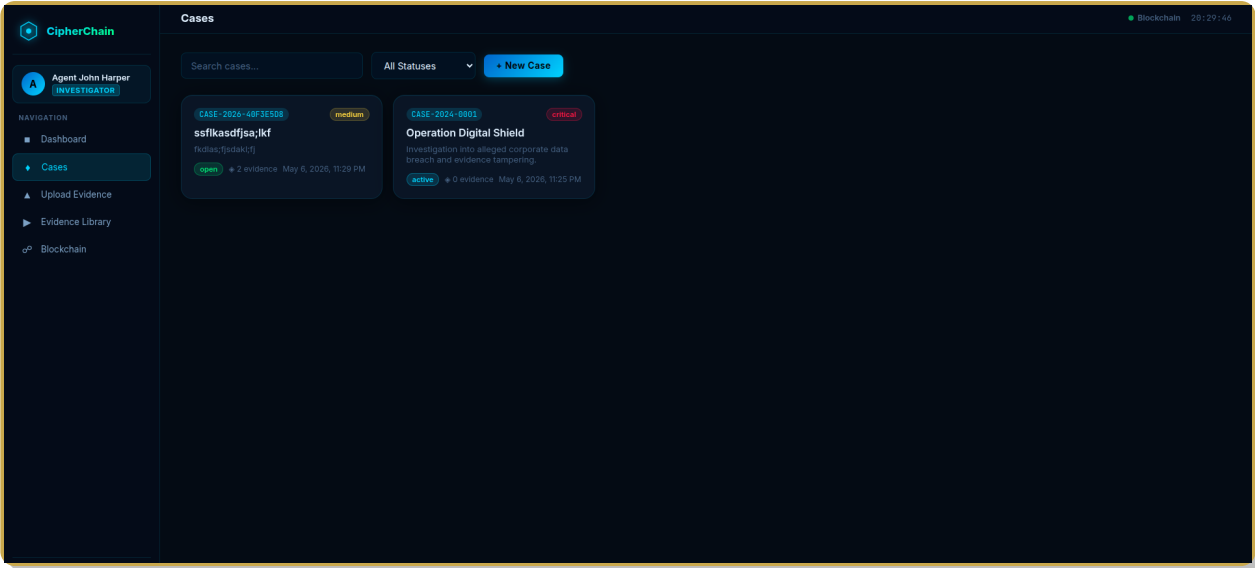
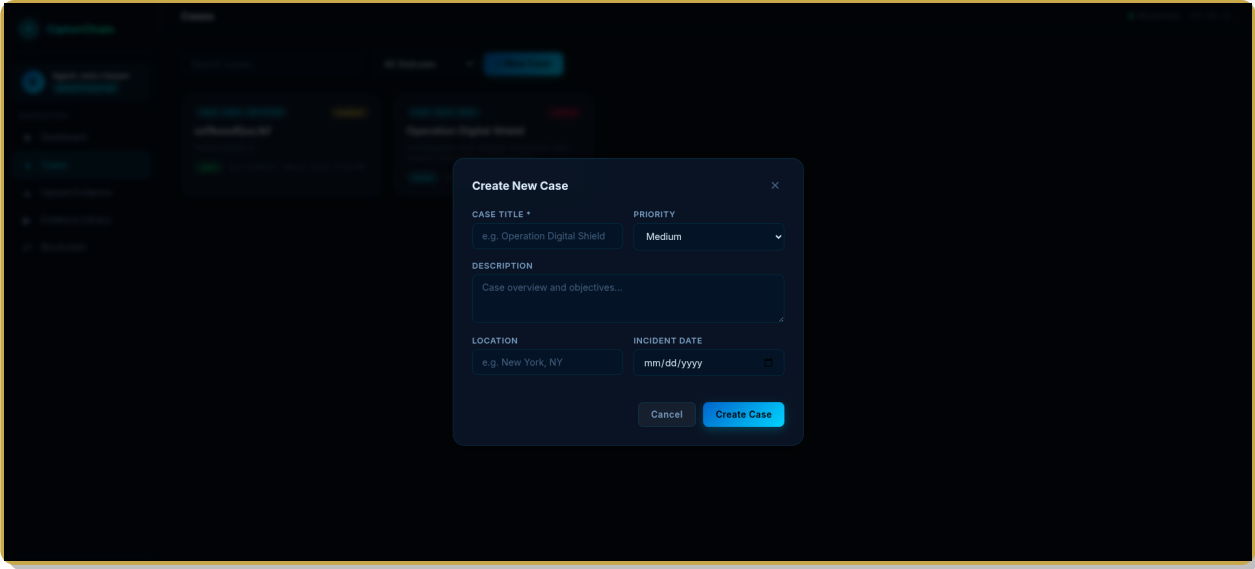
The following screenshots provide a visual overview of the Digital_V3 platform's interfaces across different roles and operational workflows.

6.1 Authentication



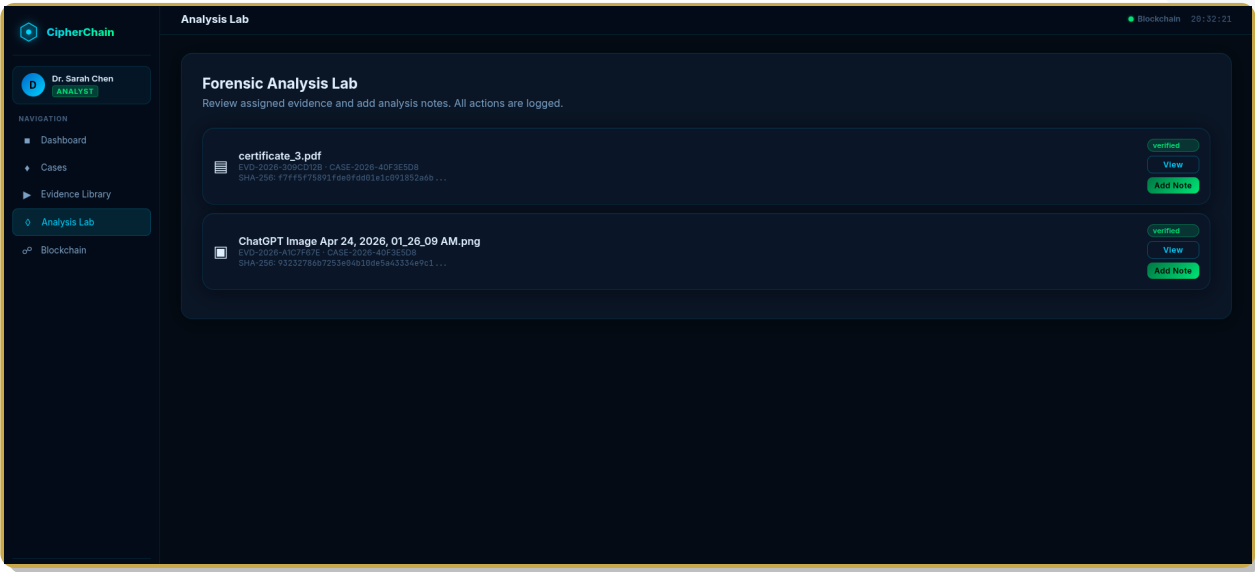
Capabilities & Workflow: The gateway to the Digital_V3 platform enforces rigorous Role-Based Access Control (RBAC). All sessions are secured using cryptographically-signed JSON Web Tokens (JWT) that prevent tampering and unauthorized access. Upon successful authentication, the system dynamically routes personnel to a specialized, icon-based dashboard tailored exclusively to their clearance level (Admin, Investigator, or Analyst), ensuring that sensitive case data remains firmly ring-fenced.

6.2 Investigator Workflow



Capabilities & Workflow: Investigators serve as the primary handlers and custodians of raw evidence. Their interface permits the creation of new forensic cases and the secure upload of seized digital artifacts. The moment a file is uploaded, the platform's forensic engine automatically computes SHA-256 and MD5 hashes, permanently anchoring the file's unique signature into the immutable custom blockchain. Concurrently, the system conducts a deep binary interrogation—extracting hidden EXIF data, GPS coordinates, and reading raw file headers—guaranteeing that the earliest stage of the chain of custody is indisputably locked in.

6.3 Analyst Workflow



Capabilities & Workflow: Analysts are granted targeted, read-only privileges restricted solely to cases explicitly assigned to them. Their dashboard provides a deep dive into the forensic microscope outputs. They can seamlessly review parsed C2PA/JUMBF manifests, evaluate calculated manipulation scores to detect AI generation or software edits, and trace file origins. Furthermore, analysts can append professional, timestamped annotations directly to the evidence’s continuous custody log, enabling collaborative investigation without ever risking the integrity of the original source artifacts.

6.4 Administrator Operations



CipherChain							
Audit Trail							
System Administrator ADMIN	May 6, 2026, 11:51 PM	System	LOGIN SUCCESS	user #1	127.0.0.1	INFO	—
	May 6, 2026, 11:51 PM	Investigator1	EVIDENCE UPLOADED	evidence #2	127.0.0.1	INFO	EVD-2026-AIC7F67E uploaded to case #2 File: ChatGPT image Apr 24, 2026, 03:26:28 AM.png (1.9 MB) Type: image SHA-256: 9d322f86b72536e4... Blockchain confirmed (TX: 4c712263d9d846...)
	May 6, 2026, 11:49 PM	System	LOGIN SUCCESS	user #1	127.0.0.1	INFO	—
	May 6, 2026, 11:41 PM	System	LOGIN SUCCESS	user #2	127.0.0.1	INFO	—
	May 6, 2026, 11:39 PM	System	LOGIN SUCCESS	user #4	127.0.0.1	INFO	—
	May 6, 2026, 11:39 PM	System	LOGIN SUCCESS	user #1	127.0.0.1	INFO	—
	May 6, 2026, 11:38 PM	System	LOGIN SUCCESS	user #1	127.0.0.1	INFO	—
	May 6, 2026, 11:31 PM	System	LOGIN SUCCESS	user #1	127.0.0.1	INFO	—
	May 6, 2026, 11:31 PM	guest1	PUBLIC VERIFY FILE	evidence	127.0.0.1	INFO	Guest verified file: certificate_3.pdf Result: verified Hash: 77f5f7509f0a0d...
	May 6, 2026, 11:31 PM	System	LOGIN SUCCESS	user #4	127.0.0.1	INFO	—
	May 6, 2026, 11:30 PM	System	LOGIN SUCCESS	user #1	127.0.0.1	INFO	—
	May 6, 2026, 11:30 PM	Investigator1	EVIDENCE UPLOADED	evidence #1	127.0.0.1	INFO	EVD-2026-309CD1D9 uploaded to case #2 File: certificate_3.pdf (26.9 KB) Type: document SHA-256: f7f5f7509f0a0d... Blockchain confirmed (TX: 76afca9f44e84f...) Desc: d8da
	May 6, 2026, 11:29 PM	Investigator1	USER ASSIGNED	case #2	127.0.0.1	INFO	User 3 - Case 2
	May 6, 2026, 11:29 PM	Investigator1	CASE CREATED	case #2	127.0.0.1	INFO	Case CASE-2026-40f3e508 created Title: softkzdfpackl Priority: medium Status: open Location: dnftrk4pdy Incident date:

Capabilities & Workflow: Administrators wield comprehensive oversight across the entire forensic pipeline to maintain operational security and compliance. Their dedicated interface allows them to provision and revoke user accounts, fine-tune role-based access configurations, and dictate global case assignments. Crucially, administrators have unimpeded access to exhaustive, system-wide audit and custody logs. They can visualize the complete timeline of any piece of evidence, track real-time custody transfers between operators, and quickly identify anomalies.

Platform Summary

Component	Technology / Capability
API & Backend	Python / Flask, RESTful, JWT-secured
Blockchain Ledger	Custom Python blockchain, append-only, hash-chained
Forensic Engine	Binary extraction: EXIF, GPS, C2PA, JUMBF, MIME, AI-detection
Database	SQLAlchemy ORM / SQLite — Users, Cases, Evidence, Audit Logs
Frontend	Vanilla SPA (HTML/CSS/JS), real-time custody visualization
Security	RBAC (Admin/Investigator/Analyst), JWT, rate limiting
Certificates	PDF court-ready reports, blockchain-referenced
Supported Evidence	Images, PDFs, Videos, Archives

Ready to Harden Your Evidence Pipeline?

Digital_V3 is available for on-premise deployment within your forensics laboratory or secure government infrastructure. No cloud dependency. No external data transfer. Complete operational sovereignty.

Contact your authorized representative for a technical demonstration and deployment scoping session.

This document is confidential and intended solely for the named client recipient. All feature descriptions reflect the Digital_V3 platform specification as of the current release version.